

Bence Ladóczki

*Researcher in Blockchain Technology and
Computer Science*

Budapest

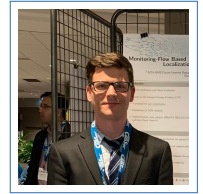
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Date of birth: 31 August 1990



Education

- 2017–2020 **Ph.D. in Computer Science**, *Kobe University*, Kobe, Japan.
Dissertation: Developments of stochastic perturbation theory and perturbative corrections of model space quantum Monte Carlo
- 2014–2017 **M.Sc. in Computer Science**, *Kobe University*, Kobe, Japan.
- 2009–2014 **B.Sc. in Electrical Engineering**, *Budapest University of Technology and Economics (BME)*, Budapest, Hungary.

Work Experience

- 2026–present **Associate Professor**, *Budapest University of Technology and Economics, VIK TMIT Department*, Budapest, Hungary.
- Research and teaching at the Department of Telecommunications and Media Informatics
 - Continuing research in blockchain technology, distributed systems, and network protocols
- 2021–2026 **Postdoctoral Researcher**, *Budapest University of Technology and Economics*, Budapest, Hungary.
- Research areas: consensus algorithms in blockchains, atomic swaps, digital signatures, layer 2 solutions, and security analysis of blockchain protocols
 - Multiple publications on Ethereum's proof-of-stake protocol, MEV auctions, and cross-chain atomic swaps
 - Contributions to vulnerability analysis of distributed randomness beacons and centralization issues in blockchain networks
- 2022–2024 **Researcher**, *Ericsson Hungary*, Budapest, Hungary.
- URLLC (Ultra-Reliable Low-Latency Communications) developments for 5G networks
 - QoE/QoS prediction for the QUIC protocol and delay-critical services in mobile networks
 - Development of machine learning models for video streaming traffic acceleration
 - Data collection framework for end-to-end radio and transport network quality monitoring
- 2023–present **Chinese Interpreter (part-time)**, *Immigration Office and National Tax Authority (NAV)*, Budapest, Hungary.
- Professional interpreting services for Chinese-speaking clients at government institutions
 - Facilitating communication in administrative and legal proceedings
- 2014–2021 **Research Assistant**, *Budapest University of Technology and Economics*, Budapest, Hungary.
- Failure localization in SDN (Software-Defined Networks)
 - Network coding in transport networks
 - Monitoring-flow based network verification
- 2013–2014 **Intern**, *Morgan Stanley*, Budapest, Hungary.
- Server-side Java implementations and GUI application programming in C#
 - Google Protocol Buffer technology and recursive tree algorithms

Teaching

- 2021–present **Course Developer and Lecturer**, *Budapest University of Technology and Economics*, Budapest, Hungary.
- Development of the curriculum and syllabus for the course “Theoretical Foundations of Cryptocurrencies” (VITMDV04)
 - Teaching blockchain technology, cryptographic principles, consensus mechanisms, and cryptocurrency protocols
 - Course topics include: digital signatures, hash functions, proof-of-work, proof-of-stake, smart contracts, and distributed ledger technology

Technical Skills

Programming languages	C, C++, Python, Go, Rust, Fortran, Java, C#
Web technologies	HTML, CSS, Node.js, React
HPC & scripting	Bash scripting, high-performance computing
Domains	Blockchain technology, quantum chemical computations, network protocols, machine learning

Languages

Hungarian	native	English: B1	Japanese: fluent	Chinese: HSK advanced	Russian: liaison interpreter (official ELTE examination)
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Research Interests

- Blockchain security and consensus mechanisms
- Distributed systems and payment networks
- Quantum chemistry and computational methods
- Machine learning applications in networks

Awards and Scholarships

2025	Best Paper Award – 7th International Conference on Blockchain Computing and Applications (BCCA), Dubrovnik, Croatia. Paper: “Predictions in MEV-Boost Auctions with Machine Learning”
2025	Distinguished Paper Award – 32nd ACM Conference on Computer and Communications Security (CCS), Taipei, Taiwan. Paper: “Forking the RANDAO: Manipulating Ethereum’s Distributed Randomness Beacon” (with Á. Nagy, J. Tapolcai, and I.A. Seres)
2014–2020	MEXT FLAGSHIP2020 Scholarship, Kobe University

Publications

Journal Articles

- 2025 **B. Ladóczki**, L. Gyevi-Nagy, P.R. Nagy, M. Kállay. "Enabling accurate and large-scale explicitly correlated CCSD(T) computations via a reduced-cost and parallel implementation." *Journal of Chemical Theory and Computation* 21(5), 2432–2447.
- 2025 A. Kotzer, **B. Ladóczki**, J. Tapolcai, O. Rottenstreich. "Addressing Scalability Issues of Blockchains with Hypergraph Payment Networks." *IEEE Transactions on Network and Service Management*.
- 2025 D. Mester, P.R. Nagy, J. Csóka, L. Gyevi-Nagy, P.B. Szabó, R.A. Horváth, et al. "Overview of developments in the MRCC program system." *The Journal of Physical Chemistry A* 129(8), 2086–2107.
- 2020 **B. Ladóczki**, M. Uejima, S.L. Ten-No. "Third-order Epstein–Nesbet perturbative correction to the initiator approximation of configuration space quantum Monte Carlo." *The Journal of Chemical Physics* 153(11).
- 2020 M. Kállay, P.R. Nagy, D. Mester, Z. Rolik, G. Samu, J. Csontos, J. Csóka, et al. "The MRCC program system: Accurate quantum chemistry from water to proteins." *The Journal of Chemical Physics* 152(7).
- 2019 **B. Ladóczki**, S.L. Ten-No. "Stochastic perturbation theory in a limited configuration space." *The Journal of Chemical Physics* 151(11).
- 2015 P. Babarczy, A. Pašić, J. Tapolcai, F. Németh, **B. Ladóczki**. "Instantaneous recovery of unicast connections in transport networks: Routing versus coding." *Computer Networks* 82, 68–80.
- 2013 Z. Rolik, L. Szegedy, I. Ladjanski, **B. Ladóczki**, M. Kállay. "An efficient linear-scaling CCSD(T) method based on local natural orbitals." *The Journal of Chemical Physics* 139(9).

Conference Papers

- 2026 **B. Ladóczki**. "Analysis of CEX-DEX Arbitrage Opportunities with Hidden Markov Models." *Proceedings of the ACM Web Conference 2026*, pp. 63–74.
- 2025 Á. Nagy, J. Tapolcai, I.A. Seres, **B. Ladóczki**. "Forking the RANDAO: Manipulating Ethereum's Distributed Randomness Beacon." *CCS '25: Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, pp. 873–887. DOI: 10.1145/3719027.3744852. (Distinguished Paper Award)
- 2025 M. Kállay, **B. Ladóczki**, P.R. Nagy. "Basis-set limit MP2 correlation energies for large molecules." *20th Central European Symposium on Theoretical Chemistry (CESTC)* (Invited talk), September 8–11.
- 2025 **B. Ladóczki**. "Accelerating Machine Learning Models for Video Streaming Traffic." *International Conference on Computing, Networking and Communications (ICNC)*.
- 2025 **B. Ladóczki**. "Predictions in MEV-Boost Auctions with Machine Learning." *7th International Conference on Blockchain Computing and Applications (BCCA)*.
- 2025 J. Tapolcai, **B. Ladóczki**. "Fully Decentralized Collection of Attestations for Single-Slot Finality in Ethereum." *IEEE 45th International Conference on Distributed Computing Systems (ICDCS)*.
- 2025 J. Tapolcai, **B. Ladóczki**, R. Lajos. "A Generic Framework for Cross-Chain Atomic Swaps of Digital Tokens." *IEEE 45th International Conference on Distributed Computing Systems (ICDCS)*.
- 2024 **B. Ladóczki**. "Mean-reverting Prices and Losses in Liquidity Pools." *6th International Conference on Blockchain Computing and Applications (BCCA)*, Dubai, United Arab Emirates.
- 2023 Z. Luo, S. Christiansson, **B. Ladóczki**, K. Komatani. "Speech Emotion Recognition using Threshold Fusion for Enhancing Audio Sensitivity." *Proceedings of the 5th ACM International Conference on Multimedia in Asia*.

- 2023 **B. Ladóczki**, Z. Luo. "Routing fee adjustment strategies in payment channels." *Fifth International Conference on Blockchain Computing and Applications (BCCA)*.
- 2023 G. Dobreff, M. Szalay, M. Molnár, L. Varga, **B. Ladóczki**, A. Báder, A. Pašić. "Predicting QoE for Delay-critical services in Mobile Networks: A video conferencing case study." *30th International Conference on Systems, Signals and Image Processing (IWSSIP)*.
- 2023 G. Dobreff, M. Szalay, **B. Ladóczki**, M. Molnár, L. Varga, A. Báder, A. Pašić. "Data collection framework for end-to-end radio and transport network quality monitoring." *15th International Conference on Quality of Multimedia Experience (QoMEX)*.
- 2022 **B. Ladóczki**, J. Bíró, J. Tapolcai. "Stochastic analysis of the success rate in atomic swaps between blockchains." *Fourth International Conference on Blockchain Computing and Applications (BCCA)*.
- 2019 **B. Ladóczki**, J. Tapolcai, A. Pašić. "Demo Abstract: Monitoring-Flow Based Network Verification and Failure Localization in SDN." *IEEE INFOCOM 2019 - IEEE Conference on Computer Communications Workshops*, Paris, France.
- 2017 **B. Ladóczki**, S.L. Ten-no. "Stochastic perturbation theories based on the MSQMC method." *20th Symposium on Theoretical Chemistry (20)*, Kyoto University, Japan.
- 2017 **B. Ladóczki**, S.L. Ten-no. "Research on the effects of truncation in perturbation theory based on stochastic methods." *11th Molecular Science Symposium (11)*, Sendai, Japan.
- 2015 **B. Ladóczki**, C. Fernandez, O. Moya, P. Babarczi, J. Tapolcai, D. Guija. "Robust network coding in transport networks." *IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, Hong Kong.

Preprints and Technical Reports

- 2026 **B. Ladóczki**, M. Rásonyi, J. Tapolcai. "Where Does MEV Really Come From? Revisiting CEX-DEX Arbitrage on Ethereum." *arXiv preprint arXiv:2604.15973*, April 17.
- 2026 M. Kállay, P.R. Nagy, **B. Ladóczki**, D. Mester. "A linear-scaling integral-direct explicitly correlated second-order Møller-Plesset approach." *ChemRxiv preprint*, February 26. DOI: 10.26434/chemrxiv.15000438
- 2025 J. Tapolcai, **B. Ladóczki**, Á. Nagy. "Slot a la carte: Centralization Issues in Ethereum's Proof-of-Stake Protocol." *Cryptology ePrint Archive*.
- 2024 B. Nagy, J. Tapolcai, **B. Ladóczki**. "Security Analysis and Vulnerabilities of TEGTSS-I Digital Signature Schemes." Technical report.

Software and Documentation

- 2026–present **B. Ladóczki**. "Luonzi: L1 Ethereum DEX Explorer." Available at: <https://luonzi.com>
- 2013–present M. Kállay, Z. Rolik, J. Csontos, I. Ladjánski, L. Szegedy, **B. Ladóczki**, et al. "MRCC: A Quantum Chemical Program Suite." Budapest University of Technology and Economics. Available at: <http://www.mrcc.hu>

Supervision

- MSc thesis Tianze Bai – "Using AI together with blockchain technology"
- Wang Fei – "Investigating DeFi exchanges with cryptography"
- MSc thesis Wang Jiayi
(2026)
- BSc thesis Benedek Zoltán; Szélyes Dávid; Soós Boglárka; Pálos Ferenc
(2026)

Academic Service

- Conferences Program Committee Member – ICDCS 2025

Publicity Chair – International Conference on Blockchain Computing and Applications (BCCA)
Reviewer – International Conference on Blockchain Computing and Applications (BCCA)
Journals Reviewer – Infocommunications Journal
Reviewer – Communications Chemistry

References

Dr. József Bíró, *Budapest University of Technology and Economics, VIK TMIT Department.*

Email: biro@tmit.bme.hu

Prof. Mihály Kállay, *Budapest University of Technology and Economics.*

Email: kallay.mihaly@vbk.bme.hu

Prof. Seiichiro L. Ten-no, *Kobe University.*

Email: tenno@garnet.kobe-u.ac.jp

Hobbies and Interests

Reading, tennis, horse riding, travelling